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#2.1

2.11.1 Problems

2.1 Consider the following five observations. You are to do all the parts of this exercise using only a calculator.

		1		2	
x	y	$x-\bar{x}$	$(x-\bar{x})^2$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
−1	1	−2	4	−1	2
0	0	−1	1	−2	2
$\sum x_i = 5$	$\sum y_i = 10$	$\sum(x_i-\bar{x}) = 0$	$\sum(x_i-\bar{x})^2$ 10	$\sum(y_i-\bar{y}) = 0$	$\sum(x_i-\bar{x})(y_i-\bar{y}) = 8$

a. Complete the entries in the table. Put the sums in the last row. What are the sample means $\bar{x}$ and $\bar{y}$?

b. Calculate b_1 and b_2 using (2.7) and (2.8) and state their interpretation.

c. Compute $\sum_{i=1}^{5} x_i^2$, $\sum_{i=1}^{5} x_i y_i$. Using these numerical values, show that $\sum(x_i-\bar{x})^2 = \sum x_i^2 - N\bar{x}^2$ and $\sum(x_i-\bar{x})(y_i-\bar{y}) = \sum x_i y_i - N\bar{x}\bar{y}$.

a. $\Sigma X_i = 3+2+1-1+0 = 5$

$\Sigma Y_i = 4+2+3+1+0 = 10$

$\bar{X} = 5 \div 5 = 1$

$\bar{Y} = 10 \div 5 = 2$

b.

$$b_2 = \frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2}$$

$$b_1 = \bar{y} - b_2\bar{x}$$

① $b_2 = 8 \div 10 = 0.8 \Rightarrow$ a one-unit increase in X_i increase y by 0.8 units on average ※

② $b_1 = 2 - 0.8 \times 1 = 1.2 \Rightarrow$ when $X=0$, the predicted value of y is 1.2 ※

c. $\sum_{i=1} X_i^2 = 3^2+2^2+1^2+(-1)^2+0^2 = 9+4+1+1 = 15$

$\sum_{i=1} X_i Y_i = 3\times4+2\times2+1\times3+(-1)\times1+0 = 12+4+3-1 = 18$

$\Sigma(X_i-\bar{X})^2 = 10 = \Sigma X_i^2 - N\cdot\bar{X}^2 = 15 - 5\times1 = 10$ ※

$\Sigma(X_i-\bar{X})(Y_i-\bar{Y}) = 8 = \Sigma X_i Y_i - N\bar{X}\bar{Y} = 18 - 5\cdot1\cdot2 = 8$ ※

d. Use the least squares estimates from part (b) to compute the fitted values of y, and complete the remainder of the table below. Put the sums in the last row.
Calculate the sample variance of y, $s_y^2 = \sum_{i=1}^{N}(y_i - \bar{y})^2/(N-1)$, the sample variance of x, $s_x^2 = \sum_{i=1}^{N}(x_i - \bar{x})^2/(N-1)$, the sample covariance between x and y, $s_{xy} = \sum_{i=1}^{N}(y_i - \bar{y})(x_i - \bar{x})/(N-1)$, the sample correlation between x and y, $r_{xy} = s_{xy}/(s_x s_y)$ and the coefficient of variation of x, $CV_x = 100(s_x/\bar{x})$. What is the median, 50th percentile, of x?

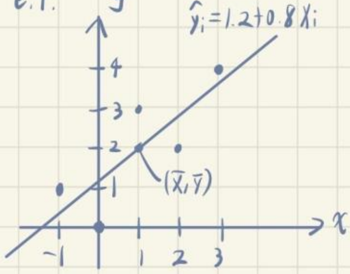
$\hat{y}_i = b_1 + b_2 x_i = 1.2 + 0.8 x_i$

$\hat{e}_i = y_i - \hat{y}_i$

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i\hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i =$ 5	$\sum y_i =$ 10	$\sum \hat{y}_i =$ 10	$\sum \hat{e}_i =$ 0	$\sum \hat{e}_i^2 =$ 3.6	$\sum x_i\hat{e}_i =$ -2

e. On graph paper, plot the data points and sketch the fitted regression line $\hat{y}_i = b_1 + b_2 x_i$.
f. On the sketch in part (e), locate the point of the means $(\bar{x}, \bar{y})$. Does your fitted line pass through that point? If not, go back to the drawing board, literally.
g. Show that for these numerical values $\bar{y} = b_1 + b_2\bar{x}$.
h. Show that for these numerical values $\bar{\hat{y}} = \bar{y}$, where $\bar{\hat{y}} = \sum \hat{y}_i/N$.
i. Compute $\hat{\sigma}^2$.
j. Compute $\widehat{\text{var}}(b_2|\mathbf{x})$ and $\text{se}(b_2)$.

e.f.



d. $s_y^2 = \sum_{i=1}^{5}(y_i - 2)^2/(5-1) = (2^2 + 0 + 1^2 + (-1)^2 + (-2)^2)/4 = 2.5$

$s_x^2 = \sum_{i=1}^{5}(X_i - 1)^2/(5-1) = (2^2 + 1^2 + 0 + (-2)^2 + (-1)^2)/4 = 2.5$

$s_{xy} = \sum_{i=1}^{5}(y_i - 2)(X_i - 1)/(5-1) = (2\times2 + 0 + 0 + (-1)(-2) + (-2)(-1))/4 = 2$

$r_{xy} = s_{xy}/s_x s_y = 2/2.5 = 0.8$

$CV = 100 \cdot \frac{S_x}{\bar{X}} = 100 \cdot \frac{\sqrt{2.5}}{1} = 158.1139$

$X_{p50} = 1$

g. $\bar{y} = 2 = b_1 + b_2\bar{x} = 1.2 + 0.8 \cdot 1 = 2$ ※

h. $\bar{\hat{y}} = \bar{y} = 2 = \frac{\sum \hat{y}_i}{N} = \frac{10}{5} = 2$ ※

i. $\hat{\sigma}^2 = \frac{\sum(y_i - \hat{y}_i)^2}{N-2} = \frac{\sum \hat{e}_i^2}{5-2} = \frac{3.6}{3} = 1.2$ ※

j. $\widehat{\text{var}}(b_2|X) = \frac{\hat{\sigma}^2}{\sum(X_i - \bar{X})^2} = \frac{1.2}{10} = 0.12$

$SE(b_2) = \sqrt{0.12} = 0.3464$

#2.22

2.22 A longitudinal experiment was conducted in Tennessee beginning in 1985 and ending in 1989. A single cohort of students was followed from kindergarten through third grade. In the experiment children were randomly assigned within schools into three types of classes: small classes with 13–17 students, regular-sized classes with 22–25 students, and regular-sized classes with a full-time teacher aide to assist the teacher. Student scores on achievement tests were recorded as well as some information about the students, teachers, and schools. Data for the kindergarten classes are contained in the data file *star5_small*. [Note: The data file *star5* contains more observations and variables.]

a. Using children who are in either a regular-sized class or a small class, estimate the regression model explaining students' combined aptitude scores as a function of class size, $TOTALSCORE_i = \beta_1 + \beta_2 SMALL_i + e_i$. Interpret the estimates. Based on this regression result, what do you conclude about the effect of class size on learning?

b. Repeat part (a) using dependent variables *READSCORE* and *MATHSCORE*. Do you observe any differences?

c. Using children who are in either a regular-sized class or a regular-sized class with a teacher aide, estimate the regression model explaining students' combined aptitude scores as a function of the presence of a teacher aide, $TOTALSCORE = \gamma_1 + \gamma_2 AIDE + e$. Interpret the estimates. Based on this regression result, what do you conclude about the effect on learning of adding a teacher aide to the classroom?

d. Repeat part (c) using dependent variables *READSCORE* and *MATHSCORE*. Do you observe any differences?

(a)

```
===============================================
                        Dependent variable:
                    ---------------------------
                            totalscore
-----------------------------------------------
small                        9.989**
                            (4.611)

Constant                    918.628***
                            (2.536)

-----------------------------------------------
Observations                  1,200
R2                            0.004
Adjusted R2                   0.003
Residual Std. Error    73.364 (df = 1198)
F Statistic          4.694** (df = 1; 1198)
===============================================
Note:               *p<0.1; **p<0.05; ***p<0.01
```

The estimated regression is $\widehat{TOTALSCORE_i} = 918.628 + 9.989\widehat{SMALL_i}$.

$\widehat{\beta_1}$=918.628→When $\widehat{SMALL_i}$=0,the predicted value of $\widehat{TOTALSCORE_i}$ is 918.628. This represents the average score for students in regular-sized class.

$\widehat{\beta_2}$=9.989→The coefficient on SMALL is statistically significant, and this means that students in small classes score about 9.989 ponits higher on average than students in regular classes.

The result suggests that smaller class sizes improve student achievement.

(b)

	Dependent variable:
	readscore
small	5.466***
	(1.897)
Constant	434.123***
	(1.044)
Observations	1,200
R2	0.007
Adjusted R2	0.006
Residual Std. Error	30.190 (df = 1198)
F Statistic	8.301*** (df = 1; 1198)
Note:	*p<0.1; **p<0.05; ***p<0.01

$$\widehat{READSCORE_i} = 434.123 + 5.466\widehat{SMALL_i}$$

	Dependent variable:
	mathscore
small	4.522
	(3.049)
Constant	484.505***
	(1.677)
Observations	1,200
R2	0.002
Adjusted R2	0.001
Residual Std. Error	48.522 (df = 1198)
F Statistic	2.199 (df = 1; 1198)
Note:	*p<0.1; **p<0.05; ***p<0.01

$$\widehat{MATHSCORE_i} = 484.505 + 4.522\widehat{SMALL_i}$$

Yes, there is a small difference. The estimated coefficient on SMALL is 5.466 for reading scores and 4.522 for math scores. This means students in small classes score higher on both subjects than those in regular classes. The effect is slightly larger for reading than for math, although the difference is not large. By the way, the coefficient on SMALL for math scores is not statistically significant, implying that the size of the class dose not have a meaningful effect on learning.

(c)

```
===============================================
                        Dependent variable:
                    ---------------------------
                            totalscore
-----------------------------------------------
aide                          -1.396
                              (4.437)

Constant                    922.145***
                              (2.640)

-----------------------------------------------
Observations                  1,200
R2                            0.0001
Adjusted R2                   -0.001
Residual Std. Error     73.504 (df = 1198)
F Statistic            0.099 (df = 1; 1198)
===============================================
Note:               *p<0.1; **p<0.05; ***p<0.01
```

$$TOT\widehat{ALSCORE}_i = 922.145 - 1.396\widehat{AIDE}_i$$

The constant (922.145) represents the average score for students in regular class without a teacher aide.

The coefficient on AIDE (-1.396) means that students in classes with a teacher aide score about 1.396 points lower on average than those in regular classes. But it is not statistically significant, implying that adding a teacher aide does not have a meaningful effect on learning.

The effect of adding a teacher aide appears to be very small and does not improve student achievement.

(d)

```
===============================================
                        Dependent variable:
                    ---------------------------
                             readscore
-----------------------------------------------
aide                          -0.372
                              (1.829)

Constant                    435.908***
                              (1.088)

-----------------------------------------------
Observations                  1,200
R2                            0.00003
Adjusted R2                   -0.001
Residual Std. Error     30.294 (df = 1198)
F Statistic            0.041 (df = 1; 1198)
===============================================
Note:               *p<0.1; **p<0.05; ***p<0.01
```

	Dependent variable:
	mathscore
aide	-1.024
	(2.931)
Constant	486.236***
	(1.744)
Observations	1,200
R2	0.0001
Adjusted R2	-0.001
Residual Std. Error	48.564 (df = 1198)
F Statistic	0.122 (df = 1; 1198)
Note:	*p<0.1; **p<0.05; ***p<0.01

$\widehat{READSCORE_i} = 435.908 - 0.372\widehat{AIDE_i}$

$\widehat{MATHSCORE_i} = 486.236 - 1.024\widehat{AIDE_i}$

Yes, there is a small difference. The estimated coefficient on AIDE is -0.372 for reading scores and -1.024 for math scores. This means students in regular classes with a teacher aide score lower on both subjects than those in regular classes. Both are not statistically significant.

#2.25

2.25 Consumer expenditure data from 2013 are contained in the file *cex5_small*. [Note: *cex5* is a larger version with more observations and variables.] Data are on three-person households consisting of a husband and wife, plus one other member, with incomes between $1000 per month to $20,000 per month. *FOODAWAY* is past quarter's food away from home expenditure per month per person, in dollars, and *INCOME* is household monthly income during past year, in $100 units.

a. Construct a histogram of *FOODAWAY* and its summary statistics. What are the mean and median values? What are the 25th and 75th percentiles?

b. What are the mean and median values of *FOODAWAY* for households including a member with an advanced degree? With a college degree member? With no advanced or college degree member?

c. Construct a histogram of ln(*FOODAWAY*) and its summary statistics. Explain why *FOODAWAY* and ln(*FOODAWAY*) have different numbers of observations.

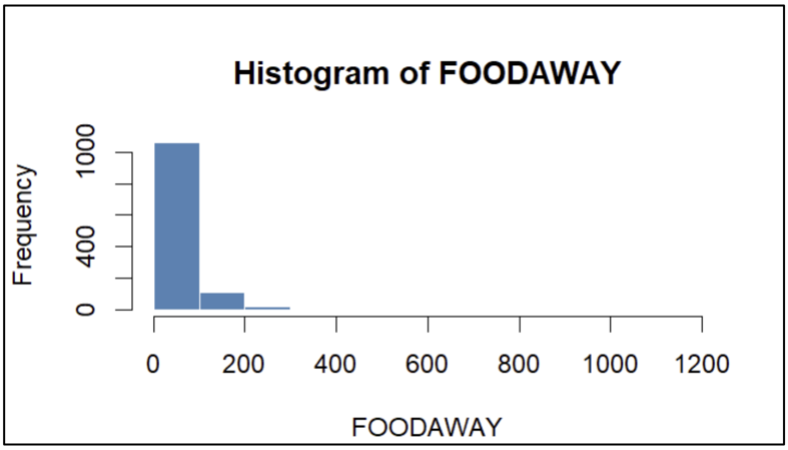
d. Estimate the linear regression $\ln(FOODAWAY) = \beta_1 + \beta_2 INCOME + e$. Interpret the estimated slope.

e. Plot ln(*FOODAWAY*) against *INCOME*, and include the fitted line from part (d).

f. Calculate the least squares residuals from the estimation in part (d). Plot them vs. *INCOME*. Do you find any unusual patterns, or do they seem completely random?

(a)

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Median	Pctl(75)	Max
foodaway	1,200	49.271	65.284	0.000	12.040	32.555	67.502	1,179.000



Statistic	Mean	Median
foodaway	73.155	48.150

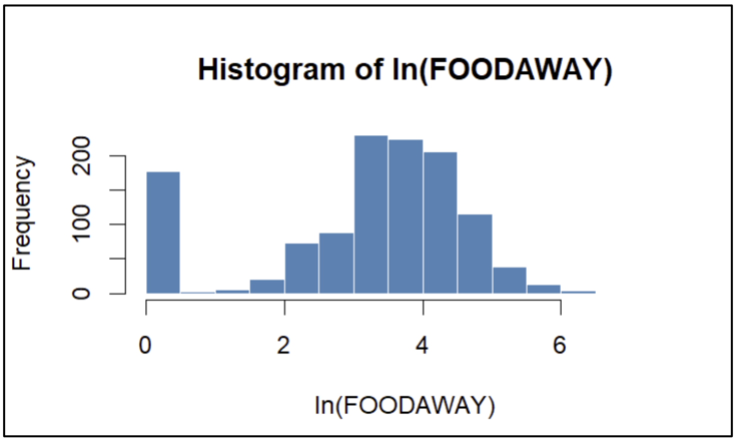
Statistic	Mean	Median
foodaway	48.597	36.110

(b)

Statistic	Mean	Median
foodaway	39.010	26.020

(c)

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Median	Pctl(75)	Max
ln_foodaway	1,200	3.143	1.544	0.000	2.568	3.513	4.227	7.073



To avoid the problem of taking the log of zero, I transformed the variable using ln(foodaway+1) instead of ln(foodaway). Some observations of FOODAWAY=0, and the log of zero is undefined. Adding 1 can ensure that the log can be computed for all observations while having only a small effect on larger values. So FOODAWAY and ln(FOODAWAY) have different numbers of observations, but adding 1 can avoid the problem of missing values.

(d)

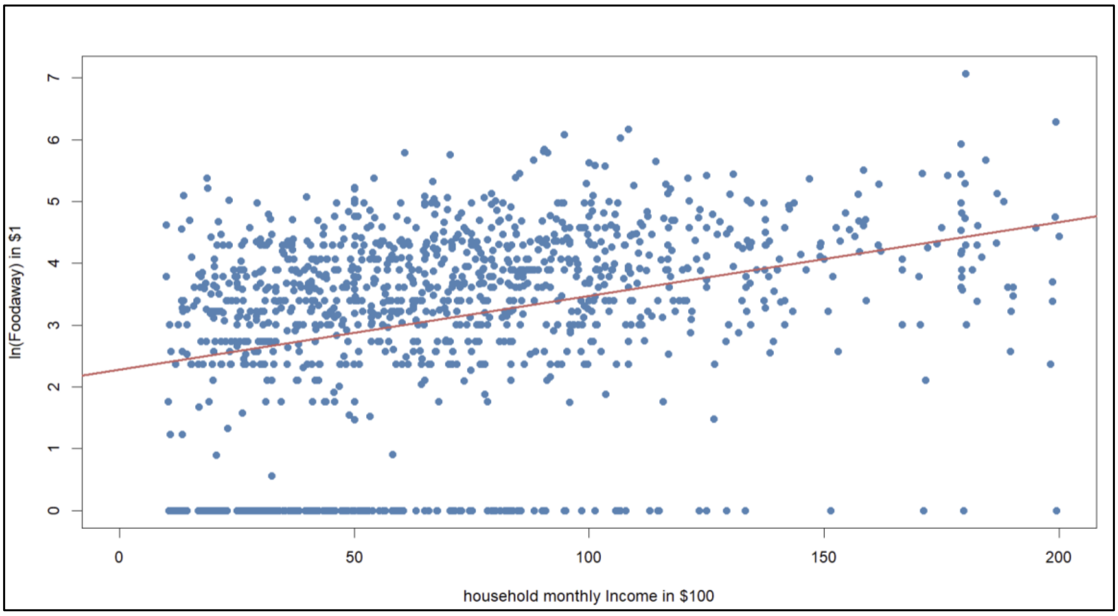
	Dependent variable:
	ln_foodaway
income	0.012***
	(0.001)
Constant	2.283***
	(0.085)
Observations	1,200
R2	0.103
Adjusted R2	0.102
Residual Std. Error	1.463 (df = 1198)
F Statistic	137.891*** (df = 1; 1198)
Note:	*p<0.1; **p<0.05; ***p<0.01

This regression uses a semi-log function form: $\widehat{\ln(FOODAWAY)}_i = 2.283 + 0.012\widehat{INCOME}$.

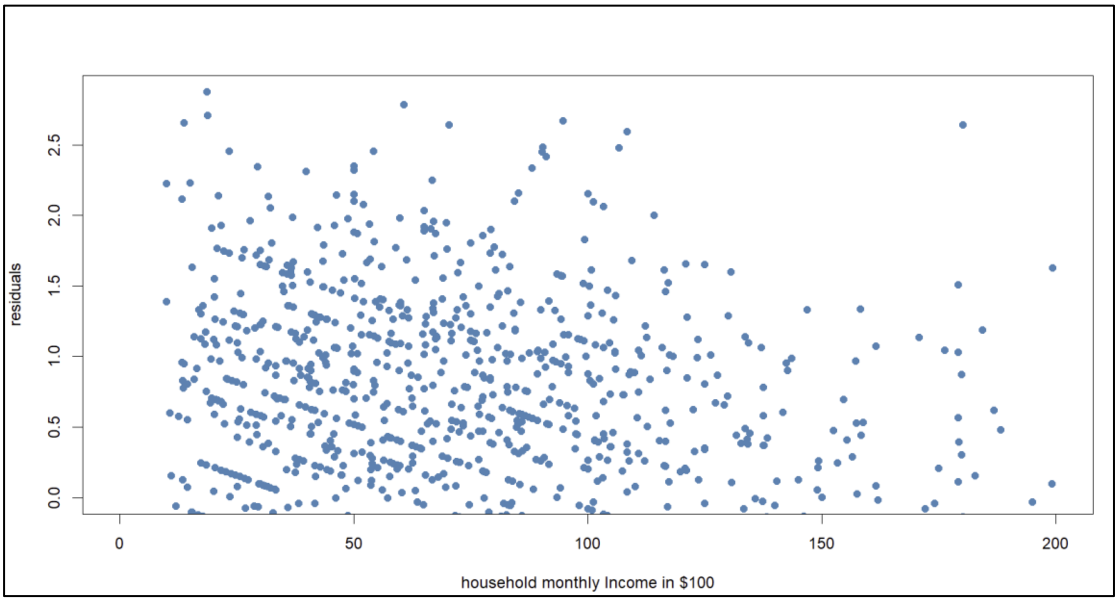
The constant (2.283) represents the predicted log value of FOODAWAY when income is zero.

The coefficient on income (0.012) means that a \$100 increase in income is associated with approximately a 1.2% increase in food away from home expenditure.

(e)



(f)



The residuals do not appear to be complete random. They exhibit heteroskedasticity. We can find that the variance of the residuals changes as the income increases, rather than staying constant(homoskedasticity).